DINESH NATARAJAN

Aerospace & Mechanical Engineer | Task Author and Evaluator for AI Training | FEA, CFD, Python

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Available 20+ hours per week with at least 4 hours of daily overlap with Pacific Time (evenings in India).

PROFILE

MSc aerospace engineer who builds expert engineering tasks for frontier-AI training and evaluation: the problem statement, a physics-based data generator with independent ground truth, the reference solution, a hardened grader and the containerised environment. On Handshake AI's Project Insight: six Hardware/CAD tasks accepted (the last two unsolved in all five frontier-model trials) and four Science tasks submitted in November 2026 after every review gate, including the combined Pass@8 difficulty gate. Built AEROTRACE, an open engineering-agent benchmark. Earlier authored CFD and structural RL tasks at Alignerr (Labelbox); now reviews aerospace solution sets at Innodata. Python, Docker, Git and Linux with ANSYS, Abaqus, HyperMesh, OpenFOAM, CadQuery/OpenCascade and MATLAB.

CORE SKILLS

- **Task and grader review:** problem statements, physics-based data generators with independent ground truth, reference solutions, automated graders (stresses, deflections, modal parameters, storey drifts, fracture toughness, mid-surface deviation), method-error measurement on 40+ validation cases, hidden evaluation cases, difficulty calibration against agent runs, bypass and reward-hacking review (sandboxed unprivileged runs, no-follow result reads, NaN and timeout probes), fail-closed numerics
- **Simulation validation:** units, equilibrium and load paths, boundary conditions, mesh-independence checks, solver health and convergence gates, reference frames, physically invalid results, time-out and runtime budgets
- **Domains:** finite-element analysis (static, modal, contact), CFD and aerodynamics (OpenFOAM, vortex lattice, airfoil analysis), structural dynamics and earthquake engineering (strong-motion records, nonlinear time-history), experimental modal analysis, fracture mechanics (ASTM E1820 J-integral), thermographic NDT, coordinate metrology (CMM), flight dynamics, fatigue and damage tolerance, B-rep geometry and GD&T (ASME Y14.5)
- **Tools:** ANSYS Mechanical, Abaqus/Standard, Altair HyperMesh, OpenFOAM, XFLR5 (XFoil), AeroSandbox, OpenSeesPy, CadQuery and OCP (OpenCascade), PTC Creo, CATIA V5, MATLAB/Simulink
- **Software practice:** Python (NumPy, SciPy, pandas, pytest), Bash, Linux (Ubuntu), Docker, Git and GitHub pull requests, CI/CD gates, Harbor task format, AI coding agents (Claude Code, Codex)

PROFESSIONAL EXPERIENCE

Task Author, Hardware/CAD and Science (Contract)
Sep 2026 to Present

Handshake AI, Project Insight | Remote

- **Hardware/CAD, six accepted tasks:** terminal-based CAD and structures tasks for a frontier-model benchmark, each through the full automated review chain (static checks, artificiality, oracle and no-op validation, implementation rubric, deep review, adversarial verifier audit, Pass@2 and Pass@5 trials); a GFEM mid-surface idealisation tool and a predictive shim-release tool were each solved in 0 of 5 trials.
- **Science, four submitted tasks:** experimental modal analysis from roving-hammer records, CMM seating of a hip-taper junction, thermographic delamination sizing in composite panels, and ASTM E1820 J-integral toughness from unloading-compliance records; each passed deep review, ambiguity and adversarial verifier audits and the combined Pass@8 gate (at most 2 of 8 frontier trials solved).
- **Exact graders and ground truth:** two-way B-rep distance with a Lipschitz bound on unsampled deviation and thickness within 2 percent; physics-based generators whose truth stays at least 1.5 bands from every decision threshold; references within two-thirds of the band on 40+ validation cases while counted shortcut methods miss by two bands or more.
- **Bypass review and hardening:** the tool under test runs as a fresh unprivileged user with resource limits, root-only verifier trees, a sealed reward channel and no-follow result reads, probed with symlinks, NaN, oversize output, fork bombs and timeouts; when a CI trial solved, traced which judgement its program handled by a shortcut, added fair traps witnessed in the public samples and re-measured it through the real verifier.

Reinforcement Learning Task Author, Engineering Simulation (Contract)
Jun 2026 to Sep 2026

Alignerr (Labelbox) | Remote

- **Reference and graders:** a verified reference solution per task scoring 1.000 against its own grader; deterministic Python graders that reject poor meshes, diverged solves and physically invalid results and score the worst case across operating conditions, including hidden evaluation cases.

- **Difficulty calibration:** retuned a wing-planform task from 0.90 to 0.62 against repeated frontier-agent runs; closed a geometry loophole; removed a ready-to-call model that let agents optimise straight to the reference; replaced an additive rubric floor that made the difficulty gate unreachable with multiplicative feasibility gating.
- **Agent run review:** diagnosed failures from transcripts (a read-only output file that blocked 5 of 5 runs; time-outs from long Monte Carlo sweeps) and reworked instructions so runs finish inside the limit (a 71-minute time-out became a 49-minute completion).
- **Portfolio and physics:** OpenFOAM wing-planform and backward-facing-step tasks behind solver-health gates; OpenSeesPy seismic retrofits with explicit dashpots in place of Rayleigh damping that over-damped under uniform ground excitation.

Aerospace Engineering Expert, AI Training (Contract) **Sep 2026 to Present**

Innodata Inc., via Upwork | Remote

- Review and rate aerospace engineering question, solution and answer sets used to train AI models, including problems built on engineering drawings; check each worked solution for technical accuracy and completeness and write a justification for every rating.

Procurement and Inventory Operations Supervisor **May 2023 to Sep 2025**

Tops | Manchester, United Kingdom

- Led a 21-member operations team running procurement, stock replenishment and inventory control; set daily priorities and tracked delivery across functions.

Product Design Engineer (Intern) **Oct 2020 to Apr 2021**

Black Space Research and Development | Bangarpet, Karnataka, India

- Built a shell-element FE model of a truck rear underrun protection device in Altair HyperMesh and solved three regulatory load cases with contact in Abaqus/Standard; identified the governing case (234.6 MPa against 230 MPa yield at the mounting flange) and recommended stronger material grades.

ENGINEERING PROJECTS

AEROTRACE: Engineering-Agent Evaluation Benchmark **2026**

Independent project | Python, JavaScript, OpenAI API | aerotrace-engineering-agent-evals.dn2066671.chatgpt.site

- Open benchmark of 28 synthetic structural-review cases in 14 trap families (draft against released revisions, unit traps, interior moment peaks, document injection), graded by a strict 8-gate grader verified against 183 of 183 injected defects, with analytical beam results cross-checked by a finite-element stiffness model; a pre-declared 168-trial study of two frontier models found two numerical errors that decision-only scoring missed.

Aircraft Engine Bracket: Topology Optimisation and Fatigue Substantiation **2023, extended 2026**

MSc dissertation, University of Salford; independent extension | PTC Creo Parametric, ANSYS Mechanical, NASGRO

- Two bracket concepts for the GE bracket challenge loads in SAE 4130; three static load cases plus modal analysis; mass cut 22 percent by topology optimisation at a factor of safety of 1.73; a mesh-independence check moved peak stress by under 1 percent; results cross-checked against hand calculations.
- 22-page substantiation: S-N, Goodman and Miner safe-life, NASGRO-equation crack growth and MIL-HDBK-1530 inspection intervals; GD&T and tolerance stack-up (worst case ±0.35 mm, RSS ±0.206 mm) in CATIA V5.

Airfoil, Flight-Dynamics and Surrogate Studies **2023 to 2025**

Independent projects | XFLR5, OpenFOAM, MATLAB/Simulink, Python

- Natural-laminar-flow comparison (NACA 63-215 against NACA 2412: peak L/D 67.3 against 46.8); Cessna 172 longitudinal state-space model meeting MIL-F-8785C short-period criteria with 6 of 6 V&V checks passed; neural-network surrogate for aerodynamic coefficients (612 samples, R^2 0.998, MAPE 2.9 percent).

EDUCATION AND CERTIFICATIONS

MSc Aerospace Engineering, University of Salford, United Kingdom, 2023 (2:1). Dissertation: structural analysis of an aircraft engine bracket by the finite element method.

BTech Aerospace Engineering, Karunya Institute of Technology and Sciences, Coimbatore, India, 2021 (2:1).

Certifications: HAI Rubrics Academy (Handshake AI, 2026); Master Diploma in Product Design (CADD Centre, 2022); Airbus Supply Chain Specialist (Airbus Beyond, 2026); Machine Learning Specialization (DeepLearning.AI and Stanford Online, 2026).